BUGFIXES

Bugfix Log

Revision: 8 **Last modified:** 2026-06-13T12:52:00Z

Per CONST-MD-Bugfix-Documentation, every bug surfaced during implementation gets a permanent entry below: title, root cause, affected files, fix description, regression guard.

Entries are append-only; do not edit historical entries except to add clarification footnotes.

2026-04-27 — boba-jackett implementation (plan 2026-04-27-jackett-management-ui-and-system-db.md)

1. Master-key two-step write silent data-loss window

Severity: HIGH (silent credential loss possible on crash mid-bootstrap).

Root cause: Initial implementation of `bootstrap.EnsureMasterKey` generated `BOBA_MASTER_KEY` in memory, returned it to the caller, and *then* wrote `.env` in a separate step. If the process crashed, was killed, or the host lost power between those two operations, the caller would proceed to encrypt credentials with a key that disk had no record of — making them permanently undecipherable.

Affected files: - `qBitTorrent-go/internal/bootstrap/bootstrap.go` - `qBitTorrent-go/internal/envfile/write.go`

Fix: Collapse the generate + write into a single `envfile.Atomic` write (`tmpfile + fsync + rename`) before the function returns. Caller never sees a key that hasn't been durably persisted. Commit `c0f20bc`.

Regression guard: - `qBitTorrent-go/internal/bootstrap/bootstrap_test.go::TestEnsureMasterKeyDoesNotDuplicateHeader` - `qBitTorrent-go/internal/envfile/write_test.go` atomic-write test layer

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2. SQLite database file mode `0644` (world-readable credentials)

Severity: CRITICAL (any local user could read encrypted credentials + ciphertext-only attack surface).

Root cause: `modernc.org/sqlite` creates database files with the process's default umask, which on the container yielded `0644` (world-readable). The Go service's `.env` was already `0600` via `envfile.Atomic`, but `boba.db` was leaking — a defense-in-depth breach since the rows are encrypted, but still a violation of the principle "the DB file should be readable only by the service owner".

Affected files: - `qBitTorrent-go/internal/db/conn.go`

Fix: After the first `Open()` succeeds, `db.Open` calls `os.Chmod(path, 0o600)` (and the `-wal` / `-shm` siblings if present), collapsing the file mode to owner-only. Commit `8405167`.

Regression guard: - Layer 4 security challenge `challenges/scripts/boba_db_file_perms_challenge.sh` - `qBitTorrent-go/internal/db/credential_leak_test.go`

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3. `AutoconfigResult` nil slices marshalled as JSON `null`

Severity: MEDIUM (silent client-breaking schema drift; UI rendered "undefined" badges instead of "0 items").

Root cause: Go's `json.Marshal` serializes a nil slice as `null`, not `[]`. When `Autoconfigure()` ran on a fresh DB with no discovered credentials, the `discovered_credentials`, `configured_now`, and `already_present` fields all marshalled as `null`. The OpenAPI 3.1 schema declared them as `array`, so clients (including the Angular dashboard) treated `null` as a contract violation.

Affected files: - `qBitTorrent-go/internal/jackett/autoconfig.go`

Fix: Pre-allocate empty slices (`[]string{}`) at the top of `Autoconfigure()` so a "nothing to do" run still serializes as `{...,"discovered":[],"configured_now":[],...}`. Commit `6f3dbaf`.

Regression guard: - Layer 6 contract test `qBitTorrent-go/tests/contract/openapi_test.go` (validates each named field is `array` per OpenAPI schema even on empty runs).

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4. Catalog `ReplaceAll` empty-input wipe risk

Severity: HIGH (would permanently wipe the indexer catalog if Jackett returned an empty response; UI would show no indexers and fuzzy matcher would have nothing to match against).

Root cause: `repos.IndexerCatalog.ReplaceAll(rows []Row)` unconditionally `DELETE`d existing rows then `INSERT`ed the new ones. If `rows` was empty (Jackett momentarily unhealthy, transient 502, or catalog parse failure), the delete still happened, leaving an empty catalog table. Next refresh would have nothing to compare against.

Affected files: - `qBitTorrent-go/internal/db/catalog.go` (or equivalent repo file)

Fix: Refuse the operation early with `errors.New("repos: ReplaceAll refusing empty replacement")`. The caller (autoconfig orchestrator) treats this as a soft error, logs it, and leaves the previous catalog intact. Commit `8d71df1`.

Regression guard: - `qBitTorrent-go/internal/jackettapi/catalog_test.go::TestRefreshCatalogAllTemplatesFailReturns200WithErrors`

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5. `TestSearchHandler_QueueFull` pre-existing flake (NOT a boba-jackett bug)

Severity: LOW (test-only; flagged for follow-up).

Status: NOT FIXED in this plan. Confirmed unrelated to boba-jackett work.

Location: `qBitTorrent-go/internal/api/api_test.go`

Symptom: Occasional intermittent failure under load when the search queue fills before the test's `wait` returns.

Action: Logged here for traceability so future audits don't mis-attribute it. Open issue suggestion: stabilise by injecting a deterministic queue-full hook instead of racing against real goroutine scheduling.

2026-06-09 — Session 7 fixes (env_loader, AsyncMock, gamestorrents)

6. env_loader flaky test: KEY2 leak across test ordering

Severity: LOW (test-only flake; not a production bug).

Root cause: `test_comment_lines_ignored` in `tests/unit/test_env_loader.py` asserted `os.environ.get("KEY2") is None` after calling `load_env_files()` with a config file where `KEY2=commented_out` was commented out. However, `load_env_files` has a "first wins" policy — if `KEY2` was already present in `os.environ` from a prior test's `load_env_files` call, the function would not override it. Under `pytest-randomly` ordering, the test that sets `KEY2` sometimes ran before the comment-line test, causing intermittent failure.

Affected files: - `tests/unit/test_env_loader.py`

Fix: Added explicit `os.environ.pop("KEY1", None)` and `os.environ.pop("KEY2", None)` at the START of the test (before calling `load_env_files`), ensuring a clean env state regardless of test execution order. The `finally` block was retained as a safety net.

Regression guard: - `tests/unit/test_env_loader.py::test_comment_lines_ignored` — ran 2147/2147 twice consecutively under random ordering with zero failures.

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7. AsyncMock warning in search deep-coverage tests

Severity: LOW (test warning; not a production bug).

Root cause: `test_iptorrents_login_no_cookies` in `tests/unit/merge_service/test_search_deep_coverage.py` used `AsyncMock()` for `login_resp` and `mock_session`. These objects are used as context managers (`async with session.post(...) as resp`), not as directly awaited coroutines. `AsyncMock.__aenter__` returns a coroutine, but the test's mock setup didn't properly wire `__aenter__`/`__aexit__`, causing "coroutine was never awaited" RuntimeWarning.

Affected files: - `tests/unit/merge_service/test_search_deep_coverage.py`

Fix: Changed `AsyncMock()` to `MagicMock()` with explicit `__aenter__` and `__aexit__` stubs, matching the actual usage pattern (context manager, not coroutine).

Regression guard: - `tests/unit/merge_service/test_search_deep_coverage.py` — 0 AsyncMock warnings from this test (was 3).

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8. gamestorrents `_parse_size` B-substring bug (documented, not fixed)

Severity: MEDIUM (plugin returns 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The `multipliers` dict in `_parse_size` iterates in insertion order: `{"B": 1, "KB": 1024, ...}`. Since `"B"` is a substring of `"KB"`, `"MB"`, `"GB"`, and `"TB"`, the check `if unit in size_str` matches `"B"` first for all realistic sizes. Then `size_str.replace("B", "")` leaves a trailing unit character (e.g. `"35.2 G"`) which `float()` cannot parse, returning 0.

Affected files: - `plugins/gamestorrents.py` (line 74-86)

Fix: NOT FIXED in this session. Documented via tests that assert actual behavior (tests suffixed `_b_substring_bug`). Fix requires reordering the dict keys to check longest units first (same approach as BOB-013).

Regression guard: - `tests/unit/test_plugin_gamestorrents.py::TestParseSize` — 8 tests documenting the bug across GB/MB/KB/TB/commas/uppercase.

2026-06-09 — Session 8 parallel plugin tests + gamestorrents fix

9. gamestorrents `_parse_size` B-substring bug (fixed)

Severity: MEDIUM (plugin returned 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The `multipliers` dict in `_parse_size` iterated in insertion order: `{"B": 1, "KB": 1024, ...}`. Since `"B"` is a substring of `"KB"`, `"MB"`, `"GB"`, and `"TB"`, the check `if unit in size_str` matched `"B"` first for all realistic sizes. Then `size_str.replace("B", "")` left a trailing unit character (e.g. `"35.2 G"`) which `float()` could not parse, returning 0.

Affected files: - `plugins/gamestorrents.py` (line 74-86)

Fix: Reordered dict keys to check longest units first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013.

Regression guard: - `tests/unit/test_plugin_gamestorrents.py::TestParseSize` — 8 tests all pass with correct byte values for GB/MB/KB/TB/B/comma/uppercase.

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10. piratebay `import os` placed after use (documented, not fixed)

Severity: LOW (only affects non-magnet torrent file downloads, which are rare).

Root cause: `piratebay.py:175` has `import os` placed after `os.fdopen` on line 172. When downloading a non-magnet torrent file, `os.fdopen` is called before `os` is imported, causing `UnboundLocalError`.

Affected files: - `plugins/piratebay.py` (line 172-175)

Fix: NOT FIXED in this session. Documented via test `test_torrent_file_download_known_bug_os_import_order`. Fix requires moving `import os` to the top of the `download_torrent` method or module level.

Regression guard: - `tests/unit/test_plugin_piratebay.py::TestDownloadTorrent::test_torrent_file_download_known_bug_os_import_order`

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11. torlock `search()` does not catch exceptions from `retrieve_url()`

Severity: LOW (network errors crash the entire search loop instead of failing gracefully).

Root cause: `torlock.py` `search()` calls `retrieve_url()` without a try/except, so a network error propagates up and crashes the search loop.

Affected files: - `plugins/torlock.py`

Fix: NOT FIXED in this session. Documented via test `test_search_exception_propagates`. Fix requires wrapping `retrieve_url()` in try/except with graceful error handling.

Regression guard: - `tests/unit/test_plugin_torlock.py::TestSearch::test_search_exception_propagates`

2026-06-09 — Session 8 wave 2: nyaa/kickass/anilibra/torrentgalaxy/yts

12. nyaa `download_torrent` missing `import re` (documented, not fixed)

Severity: MEDIUM (any nyaa.si URL passed to `download_torrent` raises `NameError`).

Root cause: `plugins/nyaa.py:170` calls `re.search()` but `import re` is absent from the module. The `search()` method works because `re` is imported transitively via other modules, but `download_torrent` fails with `NameError: name 're' is not defined`.

Affected files: - `plugins/nyaa.py` (line 170)

Fix: NOT FIXED in this session. Documented via tests that assert the `NameError`. Fix requires adding `import re` at the top of the file.

Regression guard: - `tests/unit/test_plugin_nyaa.py::TestDownloadTorrent` — 6 tests including the NameError documentation.

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13. kickass comma-separated size not matched by regex (documented, not fixed)

Severity: LOW (sizes like `1,234.5 MB` silently parse to 0).

Root cause: The HTMLParser regex `\d+\.\d+` doesn't match comma-separated numbers like `1,234.5 MB`. The comma is not handled in the regex or the replacement logic.

Affected files: - `plugins/kickass.py`

Fix: NOT FIXED in this session. Documented via test `test_comma_in_size_not_matched_by_regex`. Fix requires updating the regex to handle commas.

Regression guard: - `tests/unit/test_plugin_kickass.py::TestHTMLParserFeed::test_comma_in_size_not_matched_by_regex`

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14. kickass crash-prone patterns (fixed)

Severity: MEDIUM (empty responses from `retrieve_url` crash the plugin).

Root cause: Three methods in `kickass.py` called `retrieve_url()` without try/except or empty-response guards: `__retrieve_download_link()`, `download_torrent()`, and `search()`. When `retrieve_url` returned `None` or empty string, subsequent `re.search`/`re.sub` calls crashed with `TypeError`.

Affected files: - `plugins/kickass.py` (lines 61, 74, 94)

Fix: Added try/except + empty-response guards to all three methods. Each now handles empty/None responses gracefully (returns “NotFound”, prints fallback URL, or breaks the loop respectively).

Regression guard: - `tests/unit/test_plugin_kickass_guards.py` — 13 tests proving empty response, None response, ConnectionError, TimeoutError, and malformed HTML don’t crash.

2026-06-09 — Waves 5-8: B-substring epidemic across 9 community plugins

15. Systemic `_parse_size` B-substring bug (9 instances found, all fixed)

Severity: MEDIUM (all 9 plugins returned 0 for KB/MB/GB/TB — silently broken size reporting).

Root cause: The `_parse_size` method in each plugin uses a `multipliers` dict iterated in insertion order. With keys ordered `{"B": 1, "KB": 1024, ...}`, the check `"B" in size_str` matches the `"B"` substring inside `"KB"`, `"MB"`, `"GB"`, and `"TB"`. Then `size_str.replace("B", "")` leaves a trailing character (e.g. `"1.5 G"`) which `float()` cannot parse, returning 0.

Affected plugins (9 total): | Plugin | Status | |———|———| | `gamestorrents.py` | Fixed (BOB-024) | | `megapeer.py` | Fixed (BOB-042) | | `one337x.py` | Fixed (BOB-043) | | `extratorrent.py` | Fixed (BOB-044) | | `torrentfunk.py` | Fixed (BOB-045) | | `therarbg.py` | Fixed (BOB-047) | | `pctorrent.py` | Fixed (BOB-052, by subagent) | | `bitru.py` | Fixed (BOB-054) | | `xfsub.py` | Fixed (BOB-057) | | `yihua.py` | Fixed (BOB-058) |

Fix: Reorder dict keys longest-unit-first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013 (torrentkitty).

Global regression guard: 10 dedicated test files, each with explicit `_parse_size` assertions for all 5 units (B/KB/MB/GB/TB).

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16. Systematic `import re` omissions (3 instances found, all fixed)

Root cause: Three plugins called `re.search()` or `re.compile()` without importing `re` at module level. The `search()` method worked via transitive imports from helper modules, but `download_torrent()` failed with `NameError`.

Affected plugins: | Plugin | Status | |——–|——–| | `nyaa.py` | Fixed (BOB-035) | | `audiobookbay.py` | Fixed (BOB-042) | | (torlock `search()` also lacks exception handling — documented BOB-029) |

Fix: Added `import re` at module level.

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17. bt4g infinite loop (test-only, fixed)

Severity: LOW (test hang, not production).

Root cause: `search()` has a `while True` loop that breaks when `parser.noTorrents` is True. Tests that used `return_value=MATCHING_HTML` (not `side_effect`) caused every page to match, triggering infinite loop.

Fix: Changed to `side_effect=[MATCHING_HTML, EMPTY_HTML]` so page 2 triggers the `noTorrents` break condition.

2026-06-13 — BobaLink extension: flaky-perf-test hardening + WCAG contrast fixes

Batch verified together by `extension/ci-ext.sh` → **`CI-EXT: PASS`**, full suite **632 passed (632)**, `tsc --noEmit` clean, `npm run lint` 0/0, chrome+firefox builds loadable, §11.4.38 asset-verify pass, both store zips ≥10 KiB. Every fix below carries a permanent regression guard (§11.4.135) that was RED before the fix.

18. Junk-flood DoS security test — load-coupled absolute-wall-clock FAIL-bluff

Severity: MEDIUM (flaky test — §11.4.1 FAIL-bluff; blocked the green gate).

Root cause (FACT, not guess): `extension/tests/security/scanner-hostile-input.test.ts` asserted a tight absolute `expect(elapsed).toBeLessThan(5000)` on a 50k-anchor orchestrator scan. On a shared/oversubscribed host (load-avg 15) the *correct* scan measured **6608 ms**; in isolation it passes <5000 ms (proven 5/5). Product behaviour was always correct (exactly 2 magnets detected, all junk excluded) — only the absolute timing tripped.

Affected files: `extension/tests/security/scanner-hostile-input.test.ts`.

Fix: replaced the tight 5000 ms budget with a generous **30 s hang-ceiling** (catches a true hang / quadratic explosion — a real $O(n^2)$ over 50k anchors takes minutes) and moved the rigorous machine-independent DoS guard to a new perf test. Correctness assertions retained.

Regression guard: new `extension/tests/perf/orchestrator-scaling.perf.test.ts` asserts a metamorphic sub-quadratic scaling ratio (`t(10·N)/t(N) < 30`; linear≈10, quadratic≈100 — dimensionless, immune to host load) with a golden-good/golden-bad oracle self-validation (§11.4.107(10)): `linearRatio=9.9` accepted, `quadraticRatio=113` rejected.

19. crypto.perf p99 budget — contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): `extension/tests/perf/crypto.perf.test.ts` asserted the 80 ms budget on `dist.p99`. PBKDF2 ×2 is CPU-bound; under concurrent-suite contention p99 spiked to **164 ms** while intrinsic cost stayed ~23–41 ms (proven in isolation). The newly-added heavy parallel perf/stress suites added the contention that exposed it.

Affected files: `extension/tests/perf/crypto.perf.test.ts`.

Fix: assert the budget on `dist.min` (intrinsic, contention-robust — contention only adds time). A real ≥3.6× algorithmic regression still inflates the min past 80 ms; correctness (EXACT plaintext recovery) is asserted every round.

Regression guard: the same test (now contention-robust); isolated min ~23 ms ≤ 80; full-suite `CI-EXT: PASS`.

20. parsers.perf scaling-ratio — contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): `extension/tests/perf/parsers.perf.test.ts` computed the per-link scaling ratio from the **median** time at each size. The small (1000-link, ~5 ms) median is noise-sensitive under contention; the ratio tripped **3.09 > 3** while the intrinsic ratio is ~1.0 (proven 1.34 / 1.00 in isolation).

Affected files: `extension/tests/perf/parsers.perf.test.ts`.

Fix: compute per-link cost from the **min** run at each size (intrinsic). A genuine $O(n^2)$ regression still shows ratio ≈5 in the min run, so the cap of 3 keeps full regression-catching power while never flaking.

Regression guard: the same test; isolated ratio 1.24 ≤ 3; full-suite `CI-EXT: PASS`.

21. Three real WCAG 2.1 AA colour-contrast defects (popup theme tokens)

Severity: MEDIUM (accessibility — WCAG SC 1.4.3, real product defect).

Root cause (computed, §11.4.6): three popup theme tokens fell below the 4.5:1 normal-text contrast floor. Found by the new `extension/tests/a11y/focus-and-contrast.a11y.test.ts` (computes contrast ratios from the REAL committed CSS; analyzer self-validated against WCAG reference extremes), kept RED until fixed.

Affected files: `extension/src/popup/styles.css` (each token in both the `@media prefers-color-scheme` block and the explicit `[data-theme]` class).

Fix (same hue family, minimal visual change, ≥4.6:1): - `--text-faint` dark on `#1a1a2e`: `#7a7a90` (4.07) → `#838399` (**4.61**) - `--text-faint` light on `#f5f6fa`: `#8a8a9c` (3.14) → `#6c6c7e` (**4.76**) - `--warning-text` light on `#fff6e0`: `#9a6a00` (4.40) → `#946400` (**4.78**)

Regression guard: the 3 a11y tests flipped RED→GREEN on the fix (proving they catch the defect) and are permanent §11.4.135 guards.

2026-06-13 — BobaLink extension wave-11: options-page WCAG a11y fixes

Found by the new `extension/tests/a11y/options-contrast-motion.a11y.test.ts` (computes contrast ratios from the REAL committed `src/options/styles.css`; analyzer self-validated against WCAG reference extremes), kept RED until fixed. Batch verified by `extension/ci-ext.sh` → **`CI-EXT: PASS`** with the wave-11 coverage additions. All RED→GREEN proven; the §11.4.120 reconciliation rewrote the gates to assert the NEW mechanism (not fake-passed, not reverted).

22. Save-button text below WCAG AA over its gradient (SC 1.4.3)

Severity: MEDIUM (accessibility, real). **Root cause (computed):** `.btn-primary` has `color:#fff` over `linear-gradient(--accent → --accent-2)`; white on the lighter `--accent` (#667eea) end is only **3.66:1** (14px normal-weight text needs ≥4.5). **Fix:** the global `--accent` is ALSO `.nav-item.active` text on a dark sidebar where darkening it would REDUCE contrast, so the button uses a **button-local** darker indigo start `#5a6fce` (**4.57:1**); the `--accent-2` end is already 6.37:1. **Affected:** `src/options/styles.css .btn-primary`. **Guard:** the test now reads the button's ACTUAL gradient stops and asserts #fff ≥ 4.5 over every stop.

23 & 24. Field labels + nav-hover invisible in light theme — hard-coded literals (SC 1.4.3)

Severity: MEDIUM (accessibility, real — labels effectively unreadable in light mode). **Root cause (computed):** `.field > label` (`#d0d0e0`) and `.nav-item:hover` (`#e0e0e0`) were hard-coded **dark-theme** literals that never re-resolved per theme → **1.40:1** and **1.12:1** on the light backgrounds. **Fix:** tokenized to theme-aware tokens — label → `var(--text)` (10.6/13:1), nav-hover → `var(--text-strong)` (15.5/16.1:1) — legible in both themes. **Affected:** `src/options/styles.css`. **Guard:** the §11.4.120-reconciled test asserts the selectors now use a `var(--token)` (not a literal) AND the resolved token clears AA in BOTH themes.

25 & 26. No prefers-reduced-motion escape (SC 2.3.3 / 2.2.2)

Severity: MEDIUM (accessibility, real). **Root cause:** the options CSS ships `@keyframes fadeIn` (panel reveal on tab activation) + transitions, and the popup CSS ships transitions, but neither had a `@media (prefers-reduced-motion: reduce)` block — motion-sensitive users had no way to suppress it. **Fix:** added the standard reduced-motion block (zeroing animation/transition durations) to BOTH `src/options/styles.css` and `src/popup/styles.css`. **Guard:** the `a11y` tests assert each stylesheet ships non-trivial motion AND a reduced-motion block (RED→GREEN on the fix).

2026-06-13 — BobaLink extension wave-12: scaling-ratio FAIL-bluff hardening

Batch verified by `extension/ci-ext.sh` → **`CI-EXT: PASS`**, full suite **799 passed (799)** (+38 over wave-11: crypto-tamper 17, link-scanner-coverage 10, highlight-manager 11 — all green, no product defects). The one fix below was a flaky test exposed by the heavier 799-test load.

27. infohash dedup scaling-ratio test — single-run + tiny-floor FAIL-bluff (§11.4.50)

Severity: MEDIUM (flaky test — §11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** `tests/security/infohash-detection-hostile.test.ts`'s relative-scaling guard used a SINGLE-run `tLarge / Math.max(tSmall, 0.05)`; on a sub-ms baseline the 0.05 ms floor made the ratio noise-dominated (**53.6** observed under full-suite contention vs ~4 intrinsic). Worse, the ≤40 threshold was too loose to even catch the quadratic it guards (4× input → linear≈4, quadratic≈16 — both <40). Passed at 761 tests, tripped at 799. **Fix:** measure the MIN over 7 reps at each size (intrinsic, contention-robust — host stalls only ADD time) and tighten the threshold to **10** (sits BETWEEN linear≈4 and quadratic≈16, so it now genuinely catches an $O(n^2)$ regression while never flaking). Verified stable 5/5 in isolation + full-suite `CI-EXT: PASS`. **Affected:** `extension/tests/security/infohash-detection-hostile.test.ts`. **Note:** the sibling stress test's median estimator was initially assessed robust — but it ALSO flaked one wave later under heavier load (see entry 28), confirming that median is not enough; min is.

2026-06-13 — BobaLink extension wave-13: tab-group scaling median→min + working-tree hygiene

Batch verified by `extension/ci-ext.sh` → **`CI-EXT: PASS`**, full suite **814 passed (814)** (+15 over wave-12: `popup-states` 5 — genuine partial-Send-All gap; plus two prior-session test files brought into git — `options-save-flow` 4 + `offline-queue-persistence` 6 — that were sitting UNTRACKED in the working tree). No product defects.

28. tab-group scaling-ratio test — median estimator spikes under load (§11.4.50)

Severity: MEDIUM (flaky test — §11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** `tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts`'s scaling guard used the MEDIAN of 5 reps; under the heaviest concurrent load (814 tests) several reps landed on a busy core and dragged the median up, so the 10×-input ratio spiked past the 25 ceiling while the intrinsic ratio is ~5–8. (This refutes the wave-12 assessment that median was robust — entry 27's note.) **Fix:** switched the shared timing helper from median to **MIN** over reps (the minimum is the intrinsic-cost estimator — host stalls only ADD time, so the fastest run is closest to true cost; a genuine $O(n^2)$ regression still inflates every run including the min) and bumped reps (orchestrator 3→5, tab-group 5→9). Verified stable 3/3 in isolation (ratios 4.76/6.41/8.25 vs the 25 ceiling) + full-suite `CI-EXT: PASS`. **Affected:** `extension/tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts`. **Lesson (§11.4.118):** the robust estimator for CPU-bound scaling-ratio tests under concurrent load is MIN-of-reps, not median/mean — all six flaky-scaling fixes this session converge on it.